

CSA4302 IP

CO1 AT3

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SIMATS ENGINEERING ASSESSMENT TOOL 3

COURSE NAME: INTERNET PROGRAMMING
COURSE CODE: CSA43

COURSE OUTCOME COVERED

CO-1: Develop well-structured, standards-compliant web pages using HTML/XHTML and XML, and apply Internet protocols and HTTP communication mechanisms for web content delivery. (L2)

Assessment Tool: Code along exercise: Make your first HTML page together with CSS

Weightage: 10%

QUESTIONS:

Total Marks: 10

Scenario: Web Development Standards Implementation

A start up is developing a modern web application. During development, the team encounters multiple issues related to HTML elements, HTTP methods, XML syntax, and web communication protocols.

The following observations are made:

- The team needs to embed a promotional video on the homepage.
- A registration form must securely send user data to the server.
- The navigation menu must follow semantic HTML standards.
- XML configuration files are being used but contain syntax errors.
- The application must communicate over the correct web protocol.

Data Snippets Provided

Snippet 1: Video Embedding Options

A. `<media src="video.mp4"></media>`

B. `<video src="video.mp4" controls></video>`

1. Video Embedding

Answer:

The element is the standard HTML5 tag for embedding videos. The controls attribute provides play, pause, volume, and other controls to the user.

2. Form Submission

Answer: B – POST

POST is suitable for registration because it sends form data in the HTTP request body instead of the URL. It is more appropriate for sensitive user information and can be securely transmitted using HTTPS.

3. Navigation Element

Answer: B – <nav>

The <nav> element is the semantic HTML5 element specifically designed for navigation links. It improves accessibility, structure, and search-engine understanding of the webpage.

4. XML Syntax

Answer: B –

<note><to>User</to><from>Admin</from></note>

Option B is correct because all elements are properly opened and closed.

- **A: Missing closing </note> tag.**
- **B: Correct XML structure.**
- **C: Incorrect closing tag; <to> is closed using </from>.**
- **D: Missing > at the end of the opening tag.**

XML requires proper nesting, matching opening and closing tags, and well-formed syntax.

5. Web Communication Protocol

Answer: B – HTTP

HTTP is the standard protocol used for communication between web browsers and web servers. It enables browsers to request webpages and servers to send responses.

- FTP: Used for file transfer.**
- HTTP: Used for web communication.**
- SMTP: Used for sending email.**
- SNMP: Used for network management.**

Thus, HTTP is the correct protocol for web applications.

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